Building and room acoustics

Room acoustics is about how a room sounds from the inside: whether speech is clear and music rings. Building acoustics is about keeping sound out: whether you can hear the neighbours. At DTH the two subjects were the course “Building and room acoustics” and a project course in which the acoustics of a real building was designed. The page has an interactive reverberation calculator and a figure on sound insulation.

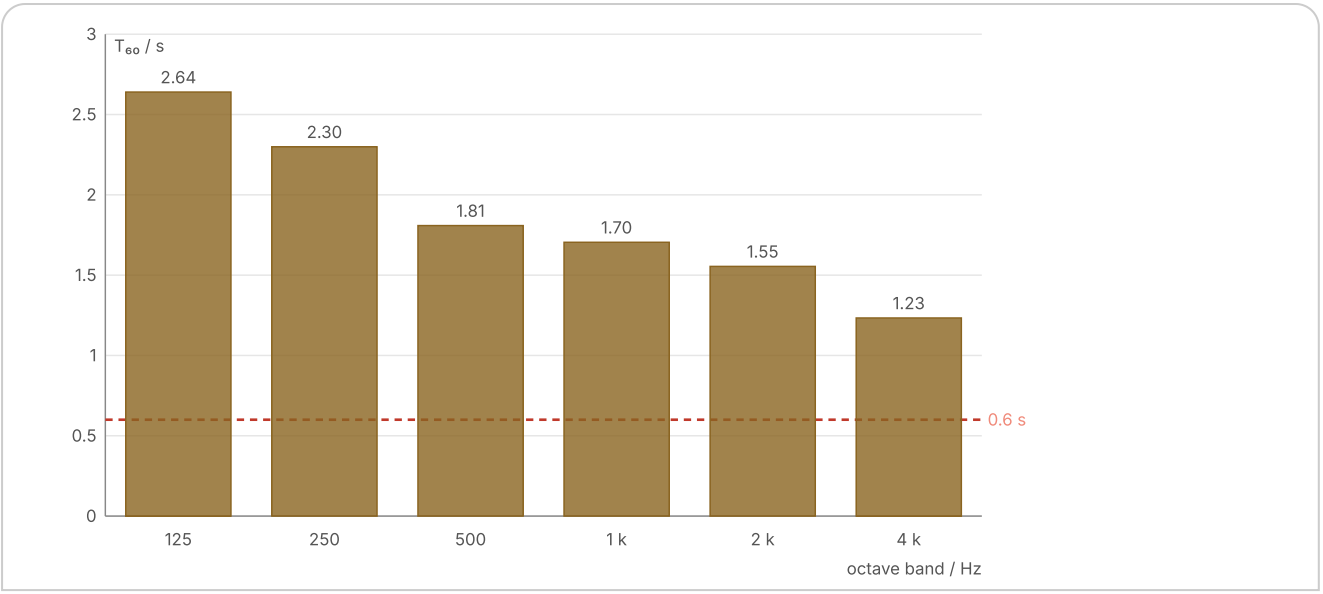
> DTH, course 5142 (1984), 6 points, and course 5147 (1988), 6 points: Building and room acoustics was one of the first courses of the degree. Four years later came “Building-acoustics design”, an integrated course in which the theory was applied to designing the acoustics of a building – from partition walls and floors to reverberation times.

Sabine and reverberation time

In 1895 the young physicist Wallace Clement Sabine was asked to do something about the new lecture hall of Harvard’s Fogg Art Museum, where nobody could understand what was said. At night he carried cushions in and out of the hall from the nearby Sanders Theatre and measured, with an organ pipe and a stopwatch, how long the tone could be heard after he stopped it. He found that the reverberation time was inversely proportional to the total absorption:

$$T_{60} = \frac{0.161\,V}{A}, \qquad A = \sum_i S_i\,\alpha_i$$

T_{60} is the time it takes the sound pressure level to fall by 60 dB, V the room volume in m³, and A the equivalent absorption area in m² (“sabins”), the sum of the surface areas times their absorption coefficients. An open window absorbs everything: $\alpha = 1$. The formula holds for a diffuse sound field and not too much absorption; Eyring improved it in 1930 to $T = 0.161V/(-S\ln(1-\bar{\alpha}))$. Sabine used the formula to design Boston Symphony Hall (1900), still counted among the world’s best concert halls. The unit of absorption bears his name.



Absorbers

Porous absorbers

Mineral wool, foam, textiles and carpets. Air is pushed back and forth in the fine pores, and friction converts sound energy into heat. They work best where the particle velocity is large – a quarter wavelength from the wall – so they are good at high frequencies and poor at low ones unless they are thick or hung with air behind them. The most important material parameter is the flow resistivity σ, and in 1970 Delany and Bazley found empirical formulas for the acoustic impedance from σ alone – the same formulas I used in my thesis.

Resonant absorbers

A Helmholtz resonator is an air cavity with a neck – like a bottle. The air in the neck oscillates as a mass on the air spring of the cavity, and the resonance frequency is $f_0 = \frac{c}{2\pi}\sqrt{\frac{S}{V\,l'}}$. Perforated panels in front of a cavity work the same way and can absorb a chosen frequency range. Panel and membrane absorbers – a thin panel in front of an air gap – absorb at low frequencies and are used to tame the bass in studios.

Room modes

A rectangular room with hard walls has natural vibrations (modes) at

$$f_{n_x n_y n_z} = \frac{c}{2}\sqrt{\left(\frac{n_x}{L_x}\right)^2 + \left(\frac{n_y}{L_y}\right)^2 + \left(\frac{n_z}{L_z}\right)^2}$$

At low frequencies the modes lie far apart, and the sound field is anything but diffuse: the bass boom of small rooms, where some notes drone and others vanish. The number of modes below the frequency f grows as $4\pi V f^3/3c^3$. Above the *Schroeder frequency* $f_S \approx 2000\sqrt{T_{60}/V}$ the modes overlap and Sabine's statistical theory holds. In a concert hall f_S is below 50 Hz; in a small listening room it can be 200 Hz. That is why a listening room should not have simple proportions like 1 : 2 : 3, which make modes coincide.

Good rooms

The right reverberation time depends on the use: around 0.4–0.6 s in classrooms and offices, 1 s in a theatre, 1.8–2.2 s in a concert hall for symphonic music and preferably 3–5 s in a church for organ music and choral singing. But reverberation time is not everything. From the 1960s Leo Beranek, Jürgen Meyer and others showed that the early reflections – the first 50–80 ms – are decisive: they reinforce the direct sound and make speech clear, while lateral reflections give music "spaciousness". This is measured with clarity indices such as

$$C_{80} = 10\log_{10}\frac{\int_0^{80\ \mathrm{ms}} p^2(t)\,dt}{\int_{80\ \mathrm{ms}}^{\infty} p^2(t)\,dt}\ \mathrm{dB}$$

The narrow, tall "shoebox" like the Musikverein in Vienna gives precisely many lateral reflections. The DR Concert Hall in Copenhagen (2009), whose acoustics were designed by Yasuhisa Toyota, is a vineyard hall where the audience sits on terraces that each provide early reflections.

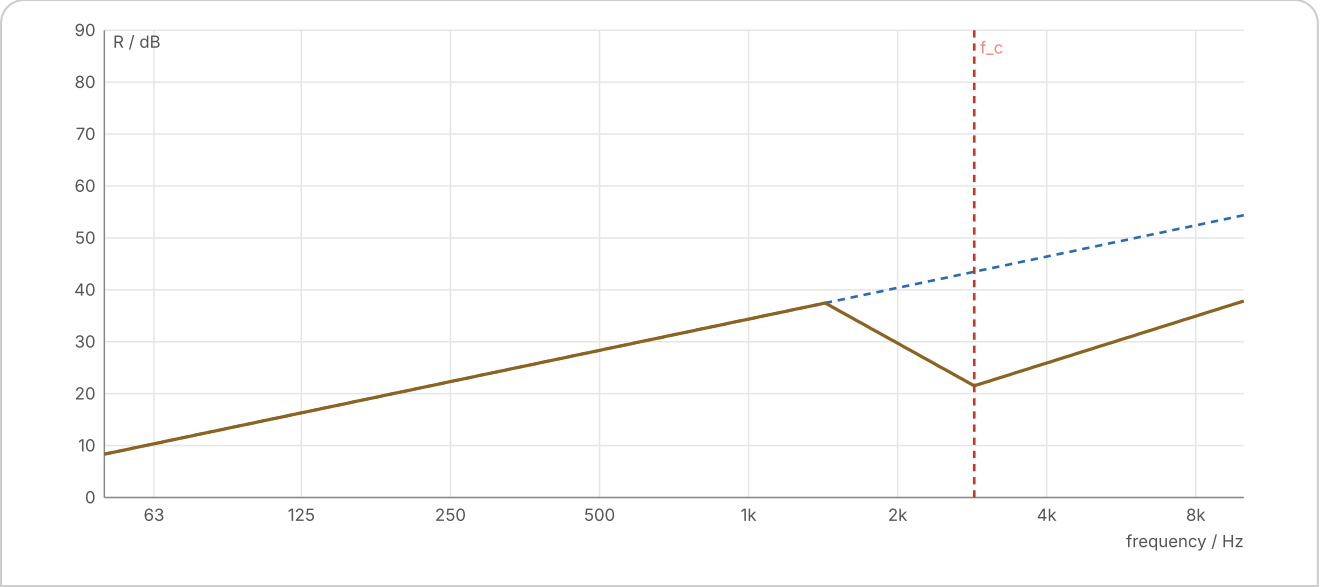
Airborne sound insulation and the mass law

The sound insulation of a wall is stated by the sound reduction index $R = 10\log_{10}(1/\tau)$, where τ is the fraction of the incident sound energy that passes through. For a single thin wall the insulation at mid frequencies is set by the mass per unit area m'': the sound pressure has to accelerate the wall, and the heavier it is, the less it moves. That gives the *mass law*:

$$R \approx 20\log_{10}(m''\,f) - 47\ \mathrm{dB} \qquad (m''\ \text{in kg/m}^2,\ f\ \text{in Hz})$$

Doubling the weight or the frequency gives 6 dB more. But a panel also has bending stiffness, and the speed of bending waves increases with frequency. At the *coincidence frequency* $f_c = \frac{c^2}{2\pi}\sqrt{\frac{m''}{B}}$ the bending wave travels as fast as the sound wave in the air along the panel, the

panel vibrates in step with the sound, and the insulation drops sharply. Thick, stiff panels have a low f_c – for 150 mm concrete it is below 150 Hz – while thin panels of plasterboard or glass have f_c around 2–3 kHz, right in the important speech frequencies.



Double walls and flanking transmission

To get high insulation without great weight, double walls are used: two panels with a cavity between them, preferably filled with mineral wool. At low frequencies the two panels and the air spring behave as a mass–spring–mass system with the resonance frequency $f_0 \approx 60/\sqrt{m'' d}$ (for m'' in kg/m² for each panel and cavity depth d in m); above it the insulation rises by 18 dB per octave instead of 6. A light partition of two layers of plasterboard on each side of a steel stud can insulate as well as a heavy brick wall.

In a finished building, however, sound does not only pass through the partition but also by detours: through the flanking walls, floor, ceiling and façade, through ventilation ducts and leaks. That is why one distinguishes the laboratory value R from the value measured in the building, R' (with a prime). Flanking paths often limit an otherwise good wall to 50–55 dB, and one of the most important tasks in design is to go through the junction details.

Impact sound

Impact sound is the sound produced when walking on a floor and heard in the room below. It is measured with a standardised tapping machine with five steel hammers falling on the floor, and stated as the impact sound pressure level L'_n in the room below – the lower the better. The most effective solution is a *floating floor*: the floor is laid on a soft, resilient layer so that it is not rigidly connected to the structural floor. Here too, one screw or skirting-board nail through the resilient layer can spoil the effect.

Building requirements

The Danish building regulations set requirements for the acoustic conditions in new buildings, and the actual values are those of sound class C in the standard DS 490. For dwellings:

Requirement	Between dwellings
Airborne sound insulation R'_w	≥ 55 dB
Impact sound level $L'_{n,w}$	≤ 53 dB
Reverberation time in stairwells	≤ 1.3 s
Noise from building services	≤ 30 dB(A) in living rooms

The subscript $_w$ means that the measurement in third-octave bands from 100 to 3150 Hz is summarised in a single number by comparison with a reference curve (ISO 717). For teaching rooms a reverberation time of at most 0.6 s is required, and less in rooms for small children.

Since 1989

- **Auralisation:** With room-acoustics simulation programs such as ODEON – developed at DTU from the mid-1980s by Jens Holger Rindel, Claus Lynge Christensen and others – you can hear a concert hall before it is built.
- **Impulse measurements:** Reverberation time and clarity are now measured with swept sines and deconvolution instead of pistol shots and interrupted noise.
- **Sound classes:** DS 490 from 2001 introduced sound classes A–D, so that better acoustic conditions than the minimum requirements can be specified.
- **Timber buildings:** Tall buildings in cross-laminated timber (CLT) are light, which brings new challenges with low-frequency impact sound.
- **Open-plan offices:** Here the issue is not long reverberation but speech intelligibility between workstations; ISO 3382-3 from 2012 provides the measures.

Further reading

- Wallace Clement Sabine: *Collected Papers on Acoustics* (1922, reprinted by Dover)
- Heinrich Kuttruff: *Room Acoustics* (6th ed., CRC 2017)
- Leo L. Beranek: *Concert Halls and Opera Houses* (2nd ed., Springer 2004)
- Jens Holger Rindel: *Sound Insulation in Buildings* (CRC 2018)

- On this site: The forest echo · Sounds from a frozen lake